

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 03
Duration: 20 Minutes

Semester : Fall25
Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:20
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1. Consider the following SDT. Draw the Annotated Parse Tree and Evaluate the values of the attributes type and width along with variable offset next to the appropriate non-terminal nodes for the following String:

```
record {
int a;
}r1;
record {
float [4] f;
}r2;
int [2] arr;
```

Afterwards draw the memory diagram along with actual location and data

Note: Offsets are given in decimal. Memory location is also given in decimal. Every memory unit is of 1 byte. The width of int and float is also given in byte in the SDT. The variable memory_addr refers to the starting position of the memory address where the variables will

$P \rightarrow \{ \text{offset} = 0; \text{memory_addr} = 58342; \} D$

$D \rightarrow T \text{ id } ; \{ \text{top.put}(\text{id.lexeme}, T.\text{type}, \text{offset}); \quad D_1$
 $\text{offset} = \text{offset} + T.\text{width}; \}$

$D \rightarrow \epsilon$

$T \rightarrow \text{record '{' } \{ \text{Env.push}(\text{top}); \text{top} = \text{new Env}(); \quad D \text{ '}' } \quad \{ T.\text{type} = \text{record}(\text{top}); T.\text{width} = \text{offset};$
 $\text{Stack.push}(\text{offset}); \text{offset} = 0; \} \quad \text{top} = \text{Env.pop}(); \text{offset} = \text{Stack.pop}(); \}$

$T \rightarrow B \{ t = B.\text{type}; w = B.\text{width}; \} C \quad \{ T.\text{type} = C.\text{type}; T.\text{width} = C.\text{width}; \}$

$B \rightarrow \text{int} \quad \{ B.\text{type} = \text{integer}; B.\text{width} = 3; \}$

$B \rightarrow \text{float} \quad \{ B.\text{type} = \text{float}; B.\text{width} = 7; \}$

$C \rightarrow \epsilon \quad \{ C.\text{type} = t; C.\text{width} = w; \}$

$C \rightarrow [\text{num}] C_1 \quad \{ C.\text{type} = \text{array}(\text{num.value}, C_1.\text{type});$
 $C.\text{width} = \text{num.value} \times C_1.\text{width}; \}$

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